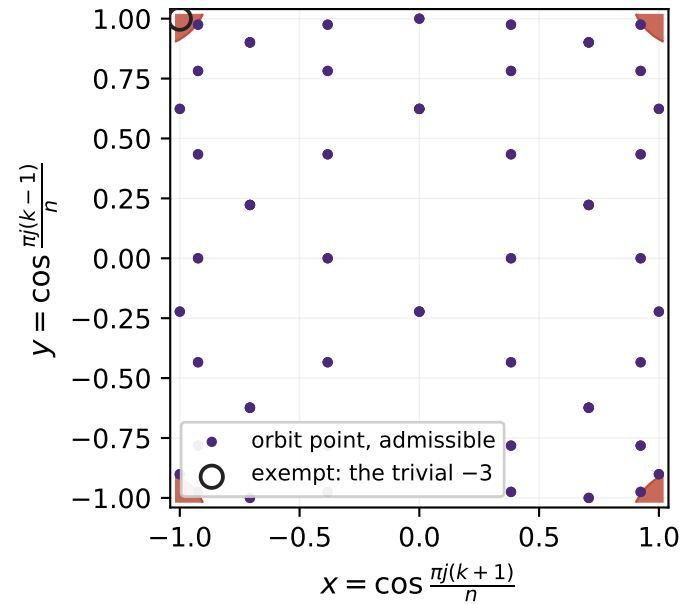
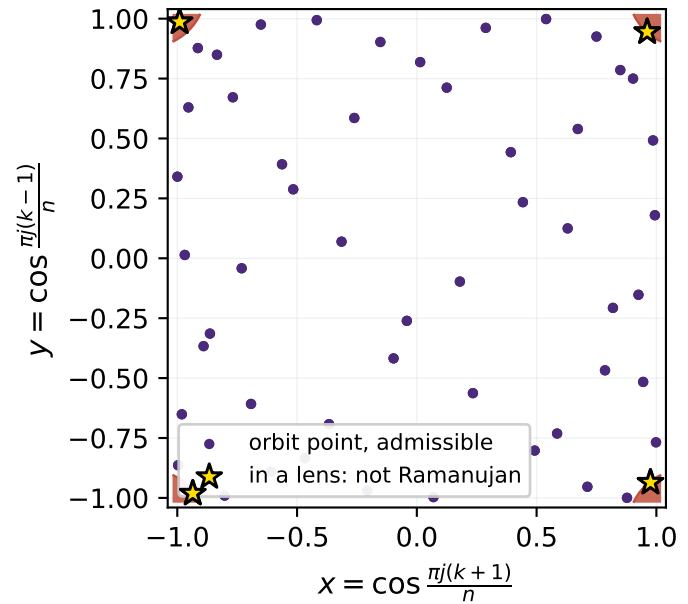


$Q(x, y) = 4x^2 + 4y^2 - 8\sqrt{2} |xy| + 3 \geq 0$
 one criterion, four fixed lenses, the same for every k

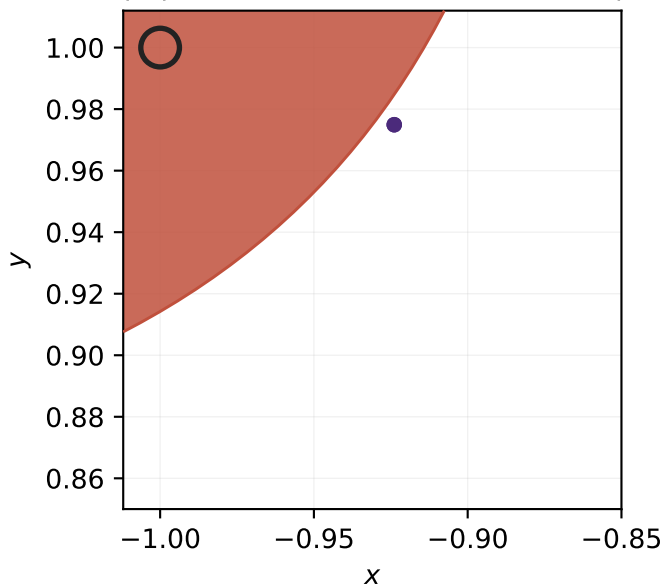
$P(112, 41)$: Ramanujan



$P(113, 41)$: not Ramanujan



zoom on the corner $(-1, +1)$
 the exempt point sits in the lens -- this is the parity law



zoom on the corner $(-1, -1)$
 8 orbit points land in a lens

